

Document Control

- Document Title: Business Requirements Document (BRD)
- System Name: Application
- Version: 1.0
- Author: Not present in source JSON
- Date: Not present in source JSON

Executive Summary

- This BRD outlines business needs for the Application, specifically ingesting customer and order CSV data into Delta tables.
- The solution will include data cleansing, computation of TotalAmount, joining logic with SCD Type 2 history, and daily customer spend aggregation, all implemented via Delta Live Tables.

Business Objectives

- Load customer and order CSV data into Delta tables.
- Cleanse data by removing null literals and duplicates.
- Compute TotalAmount in the order table.
- Join customer and order data applying SCD Type 2 for customers.
- Aggregate daily spend in a customeraggregatespend table.
- Implement the pipeline using Delta Live Tables.

In-Scope

- Source ingestion from specified file system locations.
- Creation and management of Delta tables for customer, order, ordersummary, and customeraggregatespend.
- Data cleansing according to requirements.
- Addition of computed TotalAmount.
- Implementation of SCD Type 2 logic for historical tracking.
- Aggregation for spend reporting.
- Guided by Delta Live Tables instructions.

Out-of-Scope

- Sources, systems, or fields not in source requirements.
- Security, access control, SLAs, monitoring, and alerting.
- Performance tuning and scaling (beyond stated cleansing and aggregation).
- Scheduling, orchestration, DevOps outside "Implement steps 1–10 using Delta Live Tables".
- Data quality metrics other than listed cleansing.

- Analytics/reporting UI layers.

Stakeholders

- Not present in source JSON

Business Requirements

BR-APP-001 — Source Ingestion

- Description: Load customer and order CSVs from specified file system paths into Delta tables.
- Business Value: Ensures reliable intake and storage as structured data.
- Priority: High

BR-APP-002 — Data Schemas

- Description: Define required columns for customer (CustId, Name, EmailId, Region) and order (OrderId, ItemName, PricePerUnit, Qty, Date, CustId).
- Business Value: Standardizes data for downstream processing.
- Priority: High

BR-APP-003 — Data Cleansing

- Description: Remove "Null"/Null literals and duplicate records from customer and order data.
- Business Value: Improves data quality and trustworthiness.
- Priority: High

BR-APP-004 — Computation of TotalAmount

- Description: Add computed column TotalAmount ($\text{PricePerUnit} \times \text{Qty}$) to order data.
- Business Value: Enables financial metrics for orders.
- Priority: High

BR-APP-005 — Target Table Creation: ordersummary

- Description: Create table ordersummary in gen_ai_poc_databrickscoe.sdlc_wizard as the join of customer and order data.
- Business Value: Provides unified view of customer and order info.
- Priority: High

BR-APP-006 — SCD Type 2 Customer Dimension Logic

- Description: For customer attribute changes, close past records (Inactive), add new (Active), update StartDate and EndDate, and keep all history in ordersummary.
- Business Value: Maintains compliance and historical traceability.

- Priority: High

BR-APP-007 — ordersummary Schema Enforcement

- Description: Populate ordersummary with CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date.
- Business Value: Ensures data consistency for reporting/analytics.
- Priority: High

BR-APP-008 — Spend Aggregation: customeraggregatespend

- Description: Aggregate ordersummary by Name and Date to produce customeraggregatespend with sum(TotalAmount).
- Business Value: Delivers actionable insights on customer spending.
- Priority: High

BR-APP-009 — Implementation Modality

- Description: All above requirements are implemented using Delta Live Tables as per guided steps.
- Business Value: Ensures managed and reproducible pipeline operations.
- Priority: High

Assumptions & Constraints

- No assumptions beyond provided requirements.
- All implementation to be carried out using Delta Live Tables.
- SCD tracking fields such as StartDate, EndDate, Active are logically used but not explicitly listed in schema.

Success Metrics

- All listed tables are created in Delta format as specified.
- Cleansing requirements (nulls, duplicates) are fully applied.
- ordersummary holds complete SCD Type 2 history for customer data.
- customeraggregatespend delivers correct aggregations.
- The end-to-end pipeline executes successfully using Delta Live Tables.

Risks & Mitigations

- Ambiguous field definitions for SCD Type 2 attributes (StartDate, EndDate, Active) — Mitigation: clarify in future design.
- Only basic data quality checks in scope — Mitigation: escalate if further metrics needed.
- Error handling, retries, and monitoring are not explicitly defined — Mitigation: flag for design and operational teams.

Approval

- Approver(s): Not present in source JSON

Appendix

Appendix A: Source Definitions

Source Name	Type	Location	Format	Notes
customerdata	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata	CSV	Loaded to Delta table customer
orderdata	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata	CSV	Loaded to Delta table order

Appendix B: Target Definitions

	Layer	Table	Columns	Notes
	Bronze	customer	CustId, Name, EmailId, Region	Ingested from c
	Bronze	order	OrderId, ItemName, PricePerUnit, Qty, Date, CustId	Ingested from o
	Silver	ordersummary	CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date	Joins customer
nd	Gold	customeraggregatespend	Name, TotalAmount, Date	Aggregated spe

Appendix C: Transformations

Step	Transformation	Input	Output	Logic
1	Cleansing	customer/order	customer/order	Remove "Null"/Null & duplicates
2	Computation	order	order	TotalAmount = PricePerUnit × Qty
3	Join & SCD2	customer, order	ordersummary	Join on CustId, SCD Type 2 tracking
4	Aggregation	ordersummary	customeraggregatespend	group by Name, Date; sum(TotalAmount)

Appendix D: Business Rules

Rule ID	Area	Description	Source
BR-01	Computation	TotalAmount = PricePerUnit × Qty	Requirements Input
BR-02	Cleansing	Remove null literals and duplicates	Requirements Input
BR-03	Join	Join customer–order on CustId	Requirements Input
BR-04	SCD2	Maintain SCD Type 2 (history, flags, dates)	Requirements Input
BR-05	Aggregation	Spend aggregated by Name and Date	Requirements Input

Appendix E: Process Flow

Step	Description	Dependency
1	Load customerdata to Delta table customer	Source file path
2	Load orderdata to Delta table order	Source file path

3	Apply cleansing (null/duplicate removal)	Step 1, Step 2
4	Add TotalAmount computed column	Step 2
5	Join customer with order to create ordersummary	Step 1, Step 4
6	Apply SCD Type 2 change tracking on ordersummary	Step 5
7	Create customeraggregatespend as spend aggregation	Step 6

Appendix F: Entity Relationship Diagram (ERD)

[customer]

*CustId

Name

EmailId

Region

[order]

*OrderId

ItemName

PricePerUnit

Qty

Date

CustId

TotalAmount

[ordersummary]

*CustId

Name

EmailId

Region

OrderId

ItemName

PricePerUnit

Qty

Date

StartDate

EndDate

Active

[customeraggregatespend]

*Name

TotalAmount

Date

customer *--1 order

customer *--1 ordersummary

order *--1 ordersummary

ordersummary *--* customeraggregatespend